RIPHAH
FACULTY OF
COMPUTING

Name : Ch Mubashir.

SAP : 56892.

Course : Data Visualization (BS-DS-6-1).

Instructor : Uzair Rasheed.

Date : 2-23-2026

Contents

LAB TASK 04 4
Task Information 4
Lab Purpose 4
Steps : 4
Part 1 4
Step 1 Import Libraries 4
Libraries 4
Screenshot : 5
Explanation : 5
Step 2 Loading Dataset : 5
Code : 5
Screenshot 6
Step 3 Understanding Data Structure 6
Code : 6
Screenshot : 6
Explanation : 7
Step 4 Summary Statistics : 7
Code : 7
Screenshots : 8
Explanation : 9
Part 2 9
Step 5 Visualizing Distributions : 9
5.1 Histograms (Professional Version) 9
Code : 9
Screenshots: 9
Explanation : 10
5.2 KDE Plot (Target Variable) 10
5.3 Boxplot (Outlier Detection) 11

5.4 Correlation Heatmap (Professional)..... 12
Strongest Features with SalePrice 13
Top 5 Positive Correlations..... 14
Exercise 1 – 5 Highly Skewed Variables..... 15
Exercise 2 – 3 Strongest Predictors 16

LAB TASK 04

Task Information

Lab Type: **Exploratory Data Analysis (EDA)** Part (a) - Python

Dataset: **"House Prices – Advanced Regression Techniques"**

Source: Kaggle

Lab Purpose

- Understand dataset structure
- Compute summary statistics
- Analyze distributions
- Detect missing values
- Perform correlation analysis
- Generate basic insights

Steps :

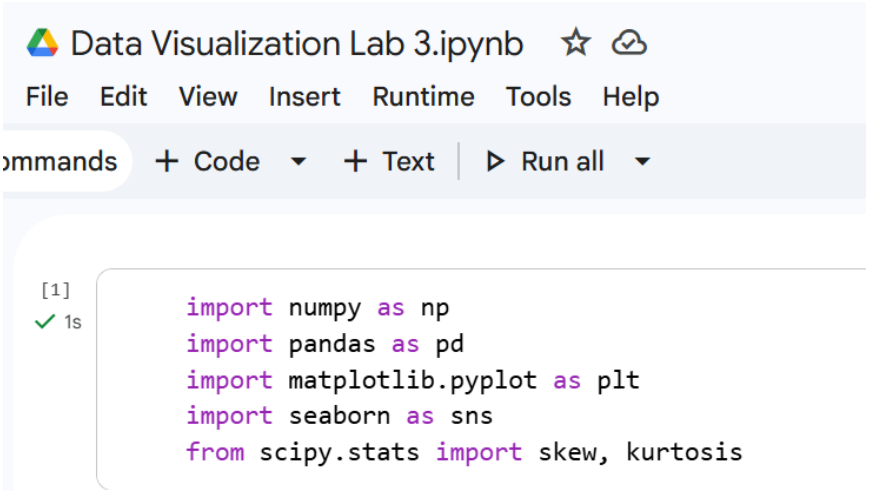
Part 1

Step 1 Import Libraries

Libraries

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns from scipy.stats

import skew, kurtosis
```

Screenshot :



Explanation :

- NumPy → numerical operations.
- Pandas → data handling.
- Matplotlib/Seaborn → visualization.
- SciPy → statistical metrics.

Step 2 Loading Dataset :

Code :

 # Load dataset

df = pd.read_csv("/content/train.csv")

 # View first 5 rows

df.head()

Screenshot

```
# Load dataset
df = pd.read_csv("/content/train.csv")

# View first 5 rows
df.head()
```

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN

5 rows × 81 columns

Toggle Gemini

iables Terminal ✓ 11:19 AM Python 3

Step 3 Understanding Data Structure

Code :

df.info() # View rows, columns, and types

print(df.isnull().sum().sort_values(ascending=False)) # Missing values

print(df.duplicated().sum()) # Duplicate check

Screenshot :

```
# Data Info
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1460 entries, 0 to 1459
Data columns (total 81 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             1460 non-null   int64
 1   MSSubClass     1460 non-null   int64
 2   MSZoning       1460 non-null   object
 3   LotFrontage    1201 non-null   float64
 4   LotArea        1460 non-null   int64
 5   Street         1460 non-null   object
 6   Alley          91 non-null     object
 7   LotShape       1460 non-null   object
 8   LandContour    1460 non-null   object
 9   Utilities      1460 non-null   object
 10  LotConfig      1460 non-null   object
 11  LandSlope      1460 non-null   object
 12  Neighborhood   1460 non-null   object
```

```
 76  MoSold         1460 non-null   int64
 77  YrSold         1460 non-null   int64
 78  SaleType       1460 non-null   object
 79  SaleCondition  1460 non-null   object
 80  SalePrice      1460 non-null   int64
dtypes: float64(3), int64(35), object(43)
memory usage: 924.0+ KB
PoolQC           1453
MiscFeature      1406
Alley            1369
Fence            1179
MasVnrType        872
                 ...
MoSold              0
YrSold              0
SaleType            0
SaleCondition       0
SalePrice           0
Length: 81, dtype: int64
0
```

Explanation :

- **Significance :** Explain that identifying missing values is the first step in data cleaning. Mention that duplicate checks ensure the integrity of the statistical analysis and prevent biased results.

Step 4 Summary Statistics :

Code :

```
#  Summary Statistics (Mean, SD, Quartiles) [cite: 39, 41, 42, 44]

print("\nSummary Statistics:\n", df.describe()) # [cite: 39]

#  Skewness & Kurtosis [cite: 45]

# Skew > 1 is highly skewed; Kurtosis > 3 has heavy tails [cite: 48, 49]

skewness = df.skew(numeric_only=True).sort_values(ascending=False) # [cite: 46]

print("\nTop Skewed Variables:\n", skewness.head(10))
```

Screenshots :

Code

[5]

```python
#  Summary Statistics (Mean, SD, Quartiles) [cite: 39, 41, 42, 44]
print("\nSummary Statistics:\n", df.describe()) # [cite: 39]

#  Skewness & Kurtosis [cite: 45]
# Skew > 1 is highly skewed; Kurtosis > 3 has heavy tails [cite: 48, 49]
skewness = df.skew(numeric_only=True).sort_values(ascending=False) # [cite: 46]
print("\nTop Skewed Variables:\n", skewness.head(10))
```

```
Missing Values:
 PoolQC           1453
```

Output :

```
Missing Values:
 PoolQC           1453
MiscFeature      1406
Alley            1369
Fence            1179
MasVnrType        872
                 ...
MoSold              0
YrSold              0
SaleType            0
SaleCondition       0
SalePrice           0
Length: 81, dtype: int64

Duplicate count: 0

Summary Statistics:
                 Id   MSSubClass  LotFrontage        LotArea  OverallQual  \
count  1460.000000  1460.000000  1201.000000    1460.000000  1460.000000
mean    730.500000    56.897260    70.049958   10516.828082     6.099315
std     421.610009    42.300571    24.284752    9981.264932     1.382997
min       1.000000    20.000000    21.000000    1300.000000     1.000000
25%     365.750000    20.000000    59.000000    7553.500000     5.000000
50%     730.500000    50.000000    69.000000    9478.500000     6.000000
75%    1095.250000    70.000000    80.000000   11601.500000     7.000000
max    1460.000000   190.000000   313.000000  215245.000000    10.000000
```

```
           PoolArea       MiscVal       MoSold       YrSold      SalePrice
count  1460.000000   1460.000000  1460.000000  1460.000000    1460.000000
mean      2.758904     43.489041     6.321918  2007.815753  180921.195890
std      40.177307    496.123024     2.703626     1.328095   79442.502883
min       0.000000      0.000000     1.000000  2006.000000   34900.000000
25%       0.000000      0.000000     5.000000  2007.000000  129975.000000
50%       0.000000      0.000000     6.000000  2008.000000  163000.000000
75%       0.000000      0.000000     8.000000  2009.000000  214000.000000
max     738.000000  15500.000000    12.000000  2010.000000  755000.000000

[8 rows x 38 columns]

Top Skewed Variables:
 MiscVal          24.476794
PoolArea         14.828374
LotArea          12.207688
3SsnPorch        10.304342
LowQualFinSF      9.011341
KitchenAbvGr      4.488397
BsmtFinSF2        4.255261
ScreenPorch       4.122214
BsmtHalfBath      4.103403
EnclosedPorch     3.089872
```

Explanation :

- **Significance :** Discuss how the SalePrice mean compares to the median. Explain that high skewness indicates the presence of outliers that might require data transformation before modeling.

Part 2

Step 5 Visualizing Distributions :

5.1 Histograms (Professional Version)

Code :

df.hist(bins=30, figsize=(18,15), color='steelblue', edgecolor='black')

plt.tight_layout()

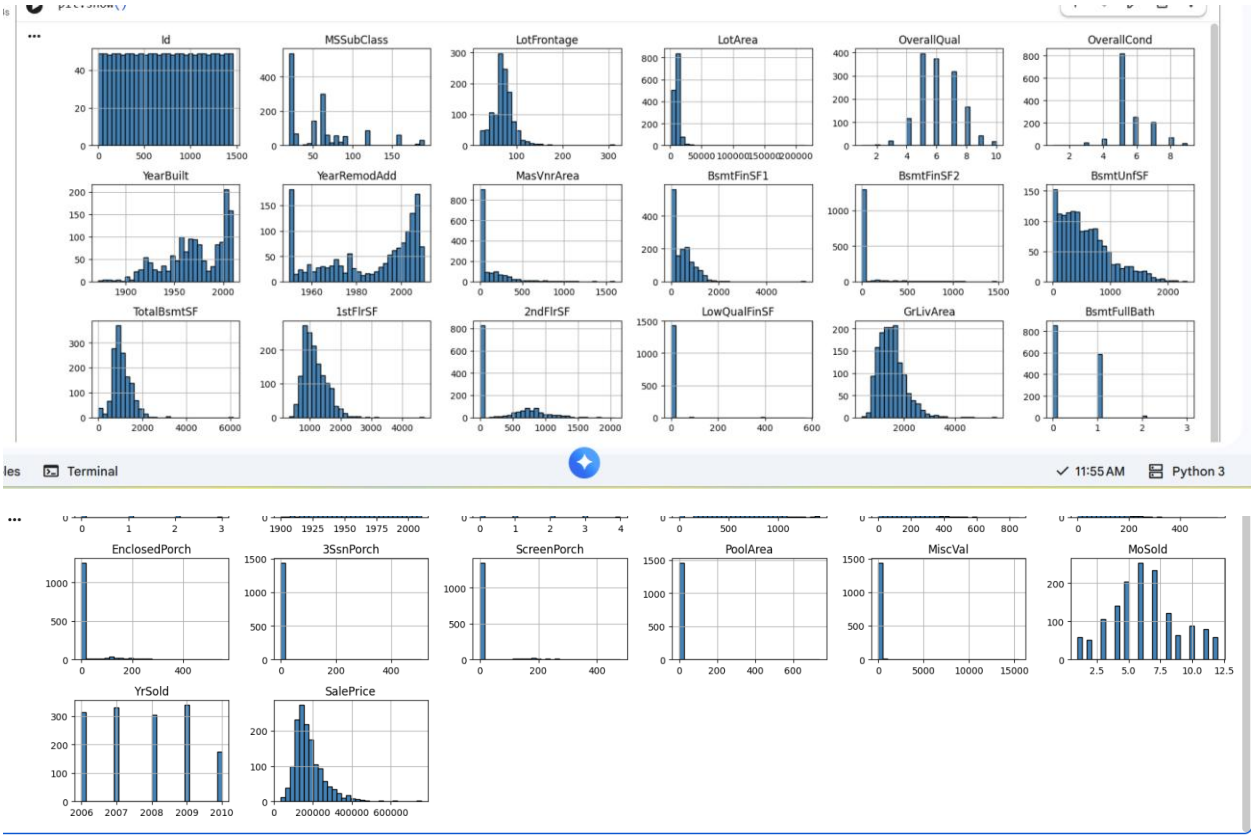
plt.show()

Screenshots:

Code :

```
df.hist(bins=30, figsize=(18,15), color='steelblue', edgecolor='black')
plt.tight_layout()
plt.show()
```

Output :



Explanation :

- **Report Insight** Several features show right-skewed distributions, particularly SalePrice and LotArea, indicating presence of high-value outliers.

5.2 KDE Plot (Target Variable)

Code :

```
plt.figure()

sns.kdeplot(df['SalePrice'], fill=True, color='darkblue')

plt.title("Distribution of SalePrice")

plt.xlabel("SalePrice")

plt.show()
```

Screenshots :

5.2 KDE Plot (Target Variable)

```
plt.figure()
sns.kdeplot(df['SalePrice'], fill=True, color='darkblue')
plt.title("Distribution of SalePrice")
plt.xlabel("SalePrice")
plt.show()
```

Interpretation

- Right skewed distribution
- High-value properties create long tail

5.3 Boxplot (Outlier Detection)

Code :

plt.figure()

sns.boxplot(x=df['SalePrice'], color='orange')

plt.title("SalePrice Boxplot")

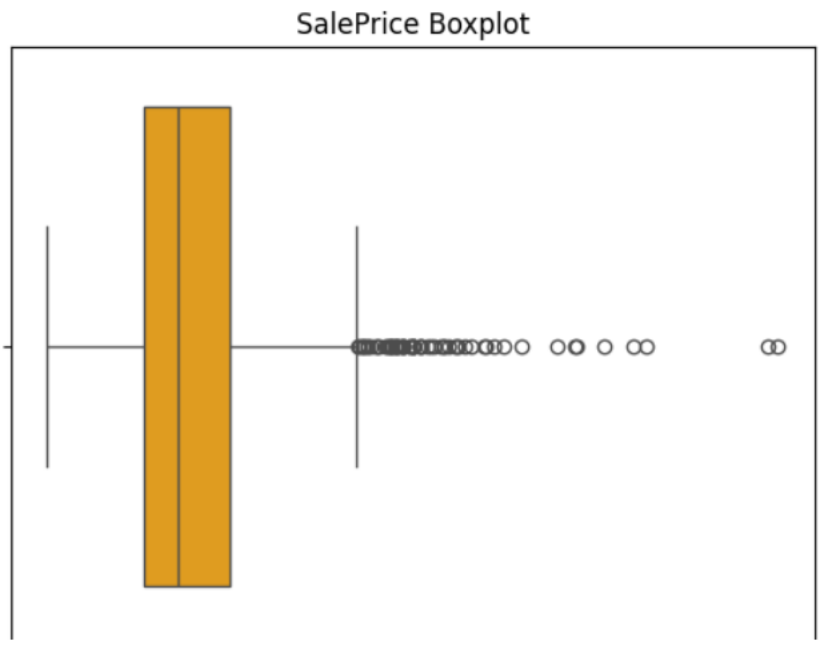
plt.show()

Screenshot

5.3 Boxplot (Outlier Detection)

[10]
✓ 0s

```
plt.figure()
sns.boxplot(x=df['SalePrice'], color='orange')
plt.title("SalePrice Boxplot")
plt.show()
```



Interpretation

- Median line
- IQR range
- Extreme upper outliers visible

5.4 Correlation Heatmap (Professional)

Code :

plt.figure(figsize=(14,12))

corr_matrix = df.corr(numeric_only=True)

sns.heatmap(corr_matrix, cmap='coolwarm', center=0)

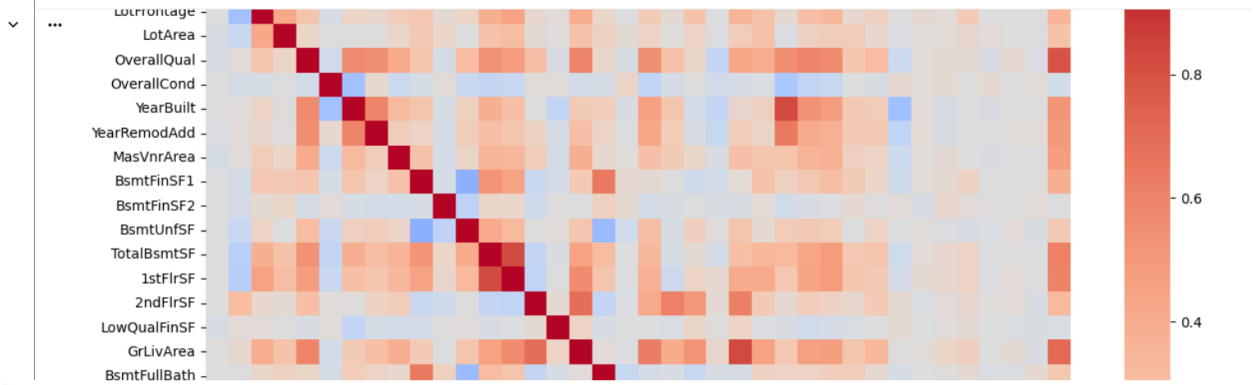
plt.title("Correlation Heatmap")

plt.show()

Screenshots:

5.4 Correlation Heatmap (Professional)

```
[11]
plt.figure(figsize=(14,12))
corr_matrix = df.corr(numeric_only=True)

sns.heatmap(corr_matrix, cmap='coolwarm', center=0)
plt.title("Correlation Heatmap")
plt.show()
```



Strongest Features with SalePrice

Code :

corr_matrix['SalePrice'].sort_values(ascending=False)

Screenshots :

Strongest Features with SalePrice

[12]
0s

```
corr_matrix['SalePrice'].sort_values(ascending=False)
```

1stFlrSF	0.605852
FullBath	0.560664
TotRmsAbvGrd	0.533723
YearBuilt	0.522897
YearRemodAdd	0.507101
GarageYrBlt	0.486362
MasVnrArea	0.477493
Fireplaces	0.466929
BsmtFinSF1	0.386420
LotFrontage	0.351799
WoodDeckSF	0.324413
2ndFlrSF	0.319334
OpenPorchSF	0.315856

Top 5 Positive Correlations

Code :

```
top_features = corr_matrix['SalePrice'].abs().sort_values(ascending=False).head(6).index

sns.pairplot(df[top_features])

plt.show()
```

Screenshots:

Exercise 1 – 5 Highly Skewed Variables

Code :

skewness[skewness > 1].head(5)

df['SalePrice_log'] = np.log1p(df['SalePrice'])

Screenshots :

Exercise 1 – 5 Highly Skewed Variables

```
skewness[skewness > 1].head(5)
```

	0
MiscVal	24.476794
PoolArea	14.828374
LotArea	12.207688
3SsnPorch	10.304342
LowQualFinSF	9.011341

dtype: float64

```
df['SalePrice_log'] = np.log1p(df['SalePrice'])
```

Suggested Transformation:

- Log transformation.
- Box-Cox transformation.

Exercise 2 – 3 Strongest Predictors

Likely:

- OverallQual
- GrLivArea
- GarageCars

Justify using:

- Correlation values
- Pairplot relationships